QP CODE: 24900268

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

Discipline Specific Core Course - MG1DSCZGY101 - BIOLOGICAL BASIS OF BEHAVIOUR 1 (2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain] [CO No(s)]

Part A. Fill in the Banks Questions
Answer all questions.
Each question carries 1 mark

1 The presence of **Na+** in the extracellular fluid contributes to the overall positive charge outside the cell. [U] [2]

2 The Goldman equation is used to calculate the **Resting Membrane Potential/ membrane potential** of a cell's membrane. [U] [2]

3 The resting membrane potential of a typical neuron is approximately **-70** mV. [U] [2]

4 The New World primates reside **South America/ Mexico** [U] [1]

5 The first known member of our genus was ***Homo habilis*** [K] [1]

6 Structures that are similar due to a common evolutionary origin are called **homologous** organs. [K] [1]

7 The **Cerebellum** receives proprioceptive input from muscles and joints to help maintain posture. [K] [3]

8 The **Cerebellum** coordinates balance, posture, and fine motor skills. [U] [3]

9 The **Limbic system** is involved in emotional responses and can influence autonomic functions. [U] [4]

10 The brain structure that regulates autonomic functions is the **Hypothalamus** [K] [4]

(10 X 1 =10 Marks)

Part B
Short Answer Type Questions
Answer 10 questions. Each question carries 2 marks

11. How does the autonomic nervous system influence emotional responses? [U] [4]

Autonomic nervous system (sympathetic and parasympathetic) regulates the emotions like facial expressions of anger, disgust, fear, happiness, sadness, or surprise are accompanied by distinct changes in heart rate, skin conductance, and skin temperature. Positive emotions increase parasympathetic nervous system activity, while negative emotions like anger increase sympathetic activity and decrease parasympathetic activity. The sympathetic nervous system (SNS) is responsible for the body's "fight-or-flight" response during stressful situations. (any two points – 2 mark)

12. Describe the relationship between the autonomic nervous system and the stress response. [U] [4]

The autonomic nervous system (ANS) plays a key role in the body's stress response by activating the sympathetic nervous system (SNS) and deactivating the parasympathetic nervous system (PSNS):

Sympathetic nervous system: This system activates the body's "fight-or-flight" response,

Parasympathetic nervous system: This system is responsible for resting.

13. Describe the two brain regions that are critical for autonomic control. [U] [4]

Hypothalamus: It coordinates the sympathetic and parasympathetic efferent pathways to regulate the body's organ systems. **Medulla oblongata**: controls autonomic functions like respiration and heartbeat

14 . What are glial cells, and how do they support the nervous system? [U] [3]

Glial cells, also known as neuroglia, are non-neuronal cells that support the nervous system by: Protecting neurons, Providing nutrients, Cleaning up, Regulating neurotransmitters Regulating blood flow and brain metabolism (Glia cell- 1 mark; Any two functions-1 mark)

15. Explain how neurotransmitters are involved in communication between neurons. [U] [3]

Neurotransmitters are the chemical messengers that transmit signals from one neuron to another, or from neurons to target cells such as muscle cells or other types of neurons. Neurotransmitters are released from the axon terminals of one neuron into the synapse, which is the small gap between the sending neuron (presynaptic neuron) and the receiving neuron (postsynaptic neuron). This release occurs in response to an electrical impulse (action potential) in the presynaptic neuron.

16. Describe the blood-brain barrier and its function. [U] [3]

The blood–brain barrier is playing a critical role in controlling the influx and efflux of biological substances essential for the brain's metabolic activity as well as neuronal function. BBB maintains the homeostasis of the brain and neuronal functions, defending the system against toxic substance, regulating the communication between the periphery and the CNS, and providing the brain with nutrients. (BBB- 1 mark; Any two functions-1 mark)

17. Mention the physical and physiological functions of neurotransmitters. [U] [2]

Neurotransmitters are chemical compounds that play a crucial role in transmitting signals in the nervous system. Physical and physiological functions: Neurotransmitters are involved in regulating many bodily functions, including: Heart rate, Breathing, Sleep cycles, Digestion, Muscle movement. (Any four points give full credit)

18. What triggers the release of neurotransmitters from synaptic vesicles? [U] [2]

Steps Triggering Neurotransmitter Release- Action Potential Arrival, Opening of Voltage-Gated Calcium Channels, Calcium Influx, Calcium Binding to Proteins, Vesicle Docking and Fusion

19. How do neurotransmitters affect target cells? [U] [2]

The response can be excitatory, inhibitory, or modulatory, depending on the neurotransmitter: (Excitatory- 1 mark; Inhibitory-1 mark)

20. Describe why it is considered acceptable to use animals in research? [U] [1]

Biological similarity, Short life cycles, Controlled environment, Safety, Benefits to humanity (Any two points give full credit)

21. Examine the main ethical concerns associated with the use of animals in scientific research. U] [1]

Animal suffering, Animal rights violations, Treatment of animals, Respect for animals. (Any two points give full credit)

22. List the animals commonly used in research. [K] [1]

Mice, Zebra Fish Frogs, Pigs, Primates, Dogs, cats etc (Any four animals give full credit)

(10 × 2 = 20 Marks)

Part C
Short Essay Type Questions
Answer 5 questions Each question carries 4 marks

23. Discuss the Na+/K+ pump and its role in maintaining the resting membrane potential [U] [2]

Definition of Na+/K+ pump.

Function: Actively transports 3 Na^+ out and 2 K^+ in against their concentration gradients using ATP.

Electrogenic Nature: Creates a net negative charge inside the cell by removing more positive ions than it brings in.

Maintains Ion Gradients: Ensures high extracellular Na^+ and high intracellular K^+, critical for membrane potential stability.

Supports Resting Potential: The pump helps maintain the resting membrane potential (~-70 mV) by counteracting ion leaks, especially K^+.

Action Potentials: Establishes ion gradients essential for nerve signaling and muscle contraction.

24. Explain the importance of the electrochemical gradient in the generation of the resting membrane potential. [U] [2]

Definition, Ion Distribution, Selective Permeability, Maintains Resting Potential

Essential for Excitability: Provides the basis for action potential generation, enabling neuron and muscle cell function. (Definition 1 mark , Explanation 3 marks)

25. How single nucleotide polymorphism influences human evolution? [U] [1]

Single nucleotide polymorphisms (SNPs) are the most common type of DNA variation in humans and are important for understanding how genetic variation contributes to human evolution, SNPs are a valuable genetic marker for studying the evolutionary history of populations. SNPs can be subject to natural selection, which can maintain or eliminate them from the gene pool. SNPs can help us understand the processes that lead to divergence between subpopulations. **(Liberal Valuation, give full marks if attempted.**

26. Write a short note on importance of natural selection [K] [1]

Natural selection is a fundamental mechanism of evolution that drives the adaptation of organisms to their environments. It acts on genetic variations within populations, favoring traits that enhance survival and reproduction. **Importance of natural selection-** Adaptation, Speciation, Fitness, Disease resistance. Definition: 1 mark Any three importance with short description 3 marks

27. How does the medulla oblongata control automatic functions and reflexes? [U] [3]

It regulates heart rate, breathing, blood pressure, and digestion.

The medulla oblongata is involved in several vital reflexes and ensure proper physiological functioning. They are Respiratory Reflexes - Coughing, Sneezing- Swallowing Reflex, Vomiting Reflex. (Any four functions: 4 mark)

28. How does the hindbrain contribute to the regulation of vital autonomic functions, such as heart rate and breathing? [U] [3]

The hindbrain plays a crucial role in regulating vital autonomic functions such as heart rate and breathing. These functions are necessary for survival and are controlled largely by the medulla oblongata and pons, which are part of the brainstem. The medulla oblongata - controlling autonomic functions like heart rate, blood pressure, and breathing. It contains vital centers that directly regulate these functions. The pons regulates breathing and acts as a bridge between the medulla and other parts of the brain involved in higher-level functions. (Any 4 points: 4 mark)

29. What protective structures surround the central nervous system, and how do they help maintain its function? [U] [4]

The central nervous system (CNS) is protected by the skull, vertebral column, meninges, and cerebrospinal fluid (CSF). Blood-Brain Barrier (BBB): Protects the brain from harmful substances in the blood while allowing essential nutrients to pass. Peripheral Nerve Sheaths: Protect peripheral nerves from injury and support their function. (Any four protective structures and its Functions- 4 marks)

(5 X 4 = 20 Marks)

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